THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 9: **The Baudhayana–Pythagoras Theorem** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 5 cm	$\sqrt{3^2+4^2}=5$
2	**(b)** 29 cm	$\sqrt{20^2+21^2}=29$
3	**(a)** 4 cm	$\sqrt{5^2-3^2}=4$
4	**(d)** 21 cm	$\sqrt{29^2-20^2}=21$
5	**(c)** Yes	$11^2+60^2 = 61^2$
6	**(b)** $4\sqrt{2}$ cm	Hypotenuse of isosceles right triangle
7	**(c)** the sum of squares on the other two sides	$c^2=a^2+b^2$
8	**(c)** $5\sqrt{2}$ cm	$\sqrt{25+25}$
9	**(a)** (5, 12, 13)	$9-4,\ 12,\ 13$
10	**(b)** 100 cm²	6^2+8^2

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	In a right triangle, the square on the hypotenuse equals the sum of the squares on the other two sides (2).	2
12	$\sqrt{81+144}=15$ cm (2).	2
13	(12, 35, 37) (2).	2
14	$100+576=676=26^2$, yes (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	By the theorem, height $=\sqrt{17^2-8^2}$ (2); $=15$ m (1).	2+1
16	By the theorem, height $=\sqrt{25^2-7^2}$ (2); $=24$ m (1).	2+1
17	By the theorem, height $=\sqrt{29^2-20^2}$ (2); $=21$ m (1).	2+1
	OR: By the theorem, height $=\sqrt{41^2-9^2}$ (2); $=40$ m (1).	2+1
18	By the theorem, height $=\sqrt{10^2-6^2}$ (2); $=8$ m (1).	2+1
	OR: By the theorem, height $=\sqrt{20^2-12^2}$ (2); $=16$ m (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Four congruent triangles in a square give $c^2=a^2+b^2$ (3). (ii) $49+576=625$ (2).	3+2
	OR: (i) 50 cm (2). (ii) Yes, exactly (1). (iii) 40 cm (2).	2+1+2
20	(i) $5\sqrt{3}\approx 8.66$ cm (3). (ii) $25\sqrt{3}\approx 43.3$ cm² (2).	3+2
	OR: (i) $x^2+(x+7)^2=(x+8)^2$ gives $x^2-2x-15=0$ (3). (ii) $x=5$: sides 5, 12, 13 (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 9: **The Baudhayana–Pythagoras Theorem** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 13 cm	$\sqrt{5^2+12^2}=13$
2	**(d)** 41 cm	$\sqrt{9^2+40^2}=41$
3	**(c)** 12 cm	$\sqrt{13^2-5^2}=12$
4	**(b)** 40 cm	$\sqrt{41^2-9^2}=40$
5	**(b)** No	$7^2+9^2\neq 11^2$
6	**(d)** $7\sqrt{2}$ cm	Hypotenuse of isosceles right triangle
7	**(c)** construct altars using rectangles and squares	Vedic geometry
8	**(d)** hypotenuse	Opposite the right angle
9	**(b)** another Pythagorean triplet	Scaling
10	**(b)** (8, 15, 17)	$64+225=289$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	5 m (2).	2
12	$6\sqrt{2}\approx 8.49$ cm (2).	2
13	Draw a square on its diagonal (2).	2
14	If $a^2+b^2=c^2$ in a triangle then it is right-angled with hypotenuse c (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	By the theorem, height $=\sqrt{25^2-15^2}$ (2); $=20$ m (1).	2+1
16	By the theorem, height $=\sqrt{26^2-10^2}$ (2); $=24$ m (1).	2+1
17	Diagonal 5 m (1.5); saved $7-5=2$ m (1.5).	1½+1½
	OR: Diagonal 13 m (1.5); saved $17-13=4$ m (1.5).	1½+1½
18	Diagonal 17 m (1.5); saved $23-17=6$ m (1.5).	1½+1½
	OR: Diagonal 25 m (1.5); saved $31-25=6$ m (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Other side 8 cm, diagonal 17 cm (3). (ii) 46 cm (2).	3+2
	OR: (i) 25 km (2). (ii) $31-25=6$ km (2). (iii) Pythagoras (1).	2+2+1
20	(i) Diagonal $\sqrt{2}$, area 2 (3). (ii) Join midpoints of the sides to get a square of half the area (2).	3+2
	OR: (i) $400+9801=10201=101^2$ (3). (ii) 990 square units (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.